BUSA8090 Assignment 1- Practical coding assignment using SQL

Submission:

- Individual submission to iLearn
- **Deadline**: 11:55 pm, Friday, 19 September 2025
- Value: 30% (Part A: 20 marks, Part B: 10 marks)

Business Background

Urban Eats Café Chain is a rapidly growing food and beverage business operating across several locations. The company offers dine in, takeaway, and delivery services. With an increasing customer base, Urban Eats wants to strengthen its data management and analytics capabilities to improve operational efficiency and maximise profitability.

Currently, Urban Eats operates 3 outlets and serves over 1,000 active customers monthly. Each outlet employs managers, chefs, servers, and baristas. Staff are assigned to a single home outlet and do not rotate between outlets; rosters are managed within that outlet. Staff can also be scheduled for multiple shifts in the same day if required. The café offers multiple product categories including beverages, meals, and desserts.

Management now seeks a structured database system that captures customer orders, staff scheduling and customer reservations to drive better decisions. The system should record shifts (date and start/end times per outlet) and keep a record of which staff worked each shift and how many hours they contributed. Customers should be able to make reservations and place orders, with payments recorded accurately.

The system should capture and organise data on outlets, staff and their shift schedules, customers, product categories and products, as well as transactions including orders, payments, and reservations.

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Task Details

Part A: **Database Design and Development** (**20 Marks**)

You are required to design and implement a relational database for Urban Eats using SQL. The system should support both day to day operations and analytical needs.

1. **Entity-Relationship (ER) Diagram**
   - Create a complete ER diagram using **Chen's** notation to represent the entities, attributes, relationships, and constraints.
   - Clearly define the primary keys, foreign keys, and cardinalities for all relationships.

2. **Database Implementation**
   - Create and implement the database using SQL.
   - Implement all necessary tables, relationships and constraints.
   - Ensure that the design reflects real world business rules for Urban Eats.

3. **Dummy Data Integration**
   - Populate the database with sample data to support data retrieval and business analytics. At Least 10 – 15 rows in each entity and up to 30 rows in key entities.
   - The dummy data should accurately reflect different business scenarios.
   - Ensure data diversity for meaningful analytics

Part B: Business Analytics Report (10 Marks)

You are acting as a Data Analyst presenting insights to Urban Eats management. The organization seeks to improve menu performance, customer loyalty, outlet efficiency, and staffing effectiveness.

Your task includes:

1. **Identifying Business Concerns:**
   Identify 5 key challenges Urban Eats faces. You may consider the following areas:
   - Sales and profitability
   - Customer retention
   - Menu optimisation
   - Staff scheduling and performance
   - Branch-level operational efficiency

2. **Formulating SQL Queries:**
   - Develop SQL queries that directly address the challenges identified.
   - Each query should be justified, explaining
     - The issue it addresses
     - How it supports management in decision making.

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3. **Comprehensive Written Report:**
   - Explain how the database was designed and implemented.
   - Discuss how SQL queries were selected and formulated.
   - Analyse the insights derived from query results and provide recommendations.
   - No word limit – ensure a clear and concise presentation.

Deliverables

- ER Diagram (Chen's notation)
- SQL scripts for database creation and data insertion
- SQL queries for business analytics
- Business analytics report in the structured format template provided **ONLY.**

This assessment evaluates your database design, implementation, and analytical skills, demonstrating how databases can support data driven business decisions.